



Lesson 1.1 — Recursive and Explicit Sequences

Big idea: Every sequence can be defined two ways: recursively (each term built from the previous) or explicitly (a formula that goes straight to the n th term).

Quick review

Recursive form tells you how to get the next term from the previous one. You must know where to start. Example: $a_1 = 3$, $a_n = a_{n-1} + 4$.

Explicit form gives you any term directly from its index n . Example: $a_n = 3 + 4(n-1)$.

When to use which: recursive is natural for word problems where each step depends on the last (interest, population). Explicit is faster when you're asked for a far-away term (like the 50th).

Worked example

Worked example. Given $a_1 = 5$ and $a_n = 2 \cdot a_{n-1} - 1$, find the first four terms.

$$a_1 = 5 \rightarrow a_2 = 2(5) - 1 = 9 \rightarrow a_3 = 2(9) - 1 = 17 \rightarrow a_4 = 2(17) - 1 = 33.$$

Warm-ups

1. Given $a_1 = 2$, $a_n = a_{n-1} + 5$, list the first five terms.
2. Given $a_n = 4n - 3$, list the first five terms.
3. Given $a_1 = 100$, $a_n = a_{n-1} \times (\frac{1}{2})$, list the first four terms.

Practice

1. A sequence starts 7, 11, 15, 19, ... Write both a recursive rule and an explicit rule.
2. A sequence is defined by $a_n = 2^n - 1$. Write an equivalent recursive rule (give a_1 and a recursion for a_n).
3. If $a_1 = 12$ and $a_n = a_{n-1} - 3$, what is a_{20} ? Answer without listing every term.

Challenge

1. A patient is prescribed 200 mg of a drug. Each day after, their body eliminates 30% of the remaining drug. Let a_n be the amount in the body on day n . Write a recursive rule and find the amount left after day 5.
2. The Fibonacci sequence is 1, 1, 2, 3, 5, 8, ... Explain why it's easy to write recursively but hard to write explicitly, in one or two sentences.



Answer key — Lesson 1.1

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 2, 7, 12, 17, 22
2. 1, 5, 9, 13, 17
3. 100, 50, 25, 12.5

Practice

1. Recursive: $a_1 = 7$, $a_n = a_{n-1} + 4$. Explicit: $a_n = 7 + 4(n-1) = 4n + 3$.
2. $a_1 = 1$; $a_n = 2 \cdot a_{n-1} + 1$.
3. $a_{20} = 12 + 19(-3) = 12 - 57 = -45$.

Challenge

1. $a_1 = 200$; $a_n = 0.70 \cdot a_{n-1}$. Day 5 $\approx 200(0.70)^4 \approx 48.02$ mg.
2. Each Fibonacci term needs the two before it, so the recursion is tiny and natural ($a_n = a_{n-1} + a_{n-2}$). An explicit formula exists (Binet's formula) but uses irrational numbers and isn't obvious from the pattern.